



THE UNIVERSITY
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Generalisation Study of Machine Learning Models for Misinformation Detection - Fake or Real

DATA7703 Application Project Report

Group No. 23

Group Members:

Neha Obilisetty (s4919561)

Remie Villatima (s4911655)

Franklin Joseph Payla (s4951623)

Akhileshwar Kolanupaka (s4930928)

Rajpriya Solanki (s4939663)

We give consent for this to be used as a teaching resource

SECTION 1 - PROBLEM

A significant social issue is the quick spread of false information online, which affects political stability, public opinion, and decisions pertaining to health. Machine learning models are now crucial tools for detecting and preventing the spread of false information due to the growing amount of digital news content. The incapacity of machine learning models to generalise across various news topics is a major limitation that has gone largely unaddressed despite tremendous advancements in fake-news classification research.

There are topic-specific patterns in a lot of popular datasets, like the Fake and Real News Dataset used in this project. Because classical machine learning models learn to recognise particular keywords, named entities, or style cues associated with topics (like political terminology), they frequently achieve very high accuracy on in-domain test sets. As a result, these models might be learning topic-related artefacts rather than the linguistic features of misinformation itself. This results in a crucial issue: A model's performance can drastically deteriorate when it is applied to a new domain (like COVID-19 news) after being trained on a different topic (like politics).

By comparing traditional machine learning models (Support Vector Machine, Logistic Regression, Random Forest, and Naïve Bayes) with a transformer-based model (DistilBERT) for fake-news detection, this project directly examines this generalisation issue. We assess whether these models overfit to the training data's subject matter or actually learn misinformation features using TF-IDF and transformer embeddings.

OBJECTIVES

The objectives of this project are:

1. To assess and compare the effectiveness of conventional machine learning models and a transformer-based model in the classification of Fake and Real news.
2. To assess the degree to which each model can generalise to unseen topics using a COVID-19 news dataset not incorporated during training.
3. To analyse whether high accuracy on the in-domain test set reflects genuine learning or merely topic memorisation.
4. To identify strengths, weaknesses, and failure modes in cross-topic fake news detection.

Research Questions

Following research questions were addressed in this project:

- Do conventional TF-IDF-based machine learning models effectively generalise to previously unseen news topics?
- Does a fine-tuned transformer model (DistilBERT) demonstrate superior generalisation compared to traditional models?
- What are the particular factors leading to generalisation failures when models face previously unencountered subjects?
- Can high in-domain performance be regarded as a dependable indicator of real-world effectiveness in misinformation detection?

SECTION 2 - SIGNIFICANCE

One of the biggest challenges in the modern society is to identify fake information. Spread of false reality directly effects public health, political stability, economic decision-making, and trust in society. The main source where fake news circulates is digital news platforms, social media. This tells us the importance of having reliable automated systems that can identify deceptive information at scale, and these systems must be effective across different topics, current and new events understanding linguistic patterns.

There has been long on-going research on fake-news classification, and approaches with performance more than 95% and in some cases 99% accuracy was also obtained. But these results are based on in-domain accuracy. However, the results can be misleading because they rarely evaluate whether models truly understand the characteristics of misinformation. Instead, models frequently exploit topic - specific correlations within the dataset, such as recurring names, political terms, or stylistic patterns. When the topic changes, these correlations disappear - exposing the model's inability to generalise.

This issue is highly significant because misinformation in the real world is dynamic. New topics emerge unexpectedly (e.g., pandemics, geopolitical conflicts, natural disasters), and fake - news classifiers must remain robust under such shifts. A model that performs exceptionally well on political news but collapses on health - related misinformation is unsuitable for real - world deployment.

By conducting a structured generalisation study, this project makes several meaningful contributions:

(1) Real - world Relevance

Our work goes beyond standard accuracy evaluation and explicitly tests whether fake - news detection models can handle unseen domains, a practical requirement for any misinformation detection system deployed by governments, media organisations, or fact - checking agencies.

(2) Addressing a Critical Research Gap

Most prior studies - especially those based on TF - IDF and classical machine learning - ignore cross - topic robustness. Transformer models such as BERT are more recent developments, but their generalisation behaviour in misinformation contexts remains underexplored. By comparing classical and transformer - based models on both in - domain and out - of - domain datasets (via the COVID - 19 subset), this project provides empirical evidence on the strengths and weaknesses of each modelling approach.

(3) Improving Model Reliability and Credibility - Evaluating generalized performance is fundamental to identify whether a model is learning the language characteristics of deception or merely retaining specific topic-related details (topic artifacts).

This improvement is crucial for building reliable AI systems that can function consistently even with genuine, real-world changes in information patterns (distribution shifts).

(4) Positive Impact on Society

Strong and Reliable misinformation detection systems can put a stop to digital deception and intentional Manipulation such as misleading vaccine information that puts lives at risk, toxic political

propaganda that can divide and confuse us, fabricated crisis reports meant to create public panic, falsified economic figures with ability to crash markets.

Therefore, deepening our comprehension of model generalization promotes the creation of more secure digital environments.

SECTION 3 - METHODOLOGY

This project followed a structured experimental pipeline consisting of data preparation, feature engineering, model training, and cross - topic generalisation testing. All experiments were conducted using Python, with the help of Scikit - Learn, Pandas, NumPy, Matplotlib, Seaborn, and Hugging Face Transformers. This section describes each component of the methodology in sufficient detail to enable full reproducibility.

1. Dataset Description

Primary Dataset (Fake/Real News Dataset - Kaggle). The main dataset consists of two files: **Fake.csv**, **True.csv**. After loading and merging these files. The count of fake articles is 23,482 and Real Articles is 21,418.

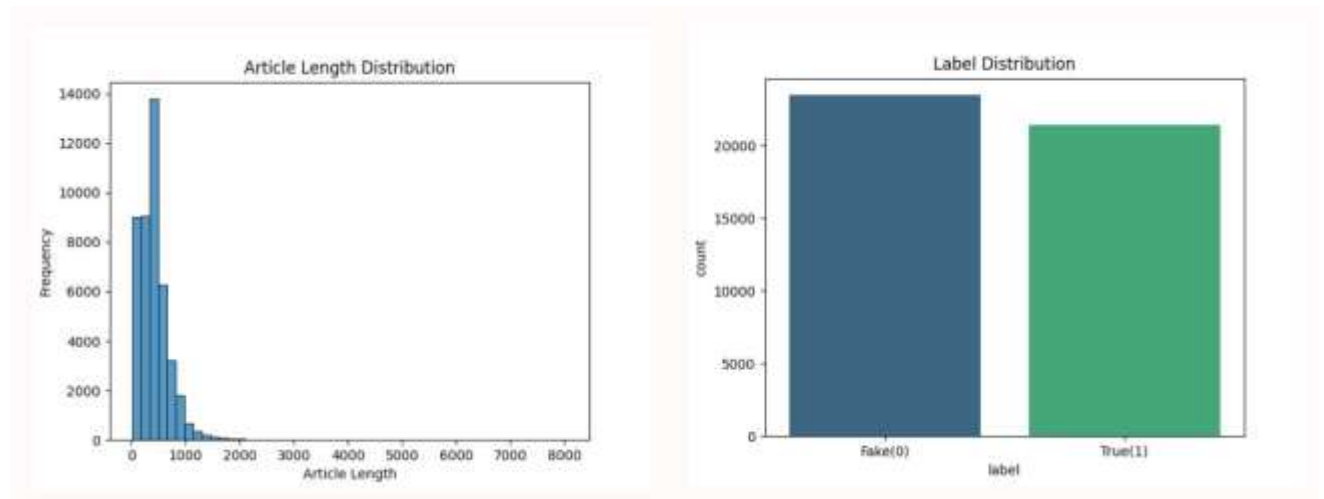


Figure 1: Article length with words distribution

Figure 2: Label distribution Fake & Real articles

Each sample contains: **title**, **text**, **subject** (politics or worldnews), **date**. A new column **label** was created: 0 = Fake, 1 = Real. A combined text field **content = title + text** was used for modelling. To evaluate topic generalisation, a third dataset was introduced:

- **NewsRealCOVID - 19.csv**: This dataset contains real news articles about COVID - 19 and was **excluded from all training**. Its purpose is to simulate a **new, real - world topic** the models have never seen.

2. Data Preprocessing

All text data underwent a cleaning process. Preprocessing was implemented using regex and the spaCy NLP library.

Steps:

1. Lowercasing
2. Removing URLs, mentions, and HTML tags
3. Removing punctuation and numeric characters
4. Collapsing multiple spaces
5. Lemmatization using spaCy's en_core_web_sm pipeline
6. Stopwords:
 - Retained negations (e.g., "not", "no") because they contribute to meaning

Duplicate and Null Handling

- 448 duplicate entries removed
- Missing values in text fields dropped (0.3% of data)

3. Feature Extraction

We implemented two feature engineering strategies

3.1. TF - IDF Vectorisation (for classical ML models)

We used Scikit - Learn's TfidfVectorizer with the following parameters: ngram_range - (1, 2), max_features - 5,000, min_df - 5, max_df - 0.9, stop_words - English, sublinear_tf - True. This produced a **5000 - dimensional sparse vector** per article.

3.2. Transformer Embeddings - DistilBERT

We used HuggingFace's pre - trained model: distilbert - base - uncased

Tokenisation used: The following parameters were used like: max length: 256, padding: max_length, truncation: True, return_attention_mask: True

Input to the model: token ID, attention masks

Output: contextual embeddings --> passed to a classification head

4. Model Training

We trained five classification models:

4.1. Logistic Regression

- Implementation: Scikit - Learn LogisticRegression
- Hyperparameters: max_iter = 1000

4.2. Naïve Bayes

- Implementation: MultinomialNB

4.3. Support Vector Machine (Linear SVM)

- Implementation: LinearSVC
- Hyperparameters: max_iter = 2000

4.4. Random Forest

- Implementation: RandomForestClassifier
- Hyperparameters: n_estimators: 200

4.5. DistilBERT (Fine - Tuned Transformer Model)

Training details: we have used already pretrained data models for DistilBERT.

Training arguments: learning_rate=2e - 5, num_train_epochs=3, weight_decay=0.01.

5. Evaluation Procedure

Train/Test Split: 80% training, 20% testing, Stratified split by label

Performance Metrics – Loss (Validation & Training, Accuracy, Precision, Recall, F1- Score

These measure both correctness and robustness.

6. Generalisation Testing

To measure real - world robustness, we evaluated trained models on an **unseen - domain COVID - 19 dataset**. This setup tests whether:

- models learned **misinformation patterns**, or
- models simply memorised **topic - specific artefacts**

Generalisation results were compared directly to in - domain test results.

SECTION 4 - FINDINGS

The experiments employing both traditional machine learning models and a transformer-based model (Distil BERT) are shown in this section. The results are separated into two categories: (1) Cross - Topic Generalisation Performance and (2) In – Domain performance.

1. In - Domain Performance (Same Topic Distribution)

An 80/20 train-test split selected from the same subject distribution as the training data (political and international news) was used to initially assess each model. Accuracy and F-1 score were used to calculate metrics.

1.1. Performance Summary

Model	Accuracy	F1 - Score
Logistic Regression	98.78%	0.9888
Naïve Bayes	94.68%	0.9511
Support Vector Machine (SVM)	99.53%	0.9956
Random Forest	99.49%	0.9953
DistilBERT (Transformer)	99.94%	0.9995

The exceptionally high performance of the SVM, Random Forest, and Distil BERT models indicates that the dataset is highly learnable by transformer models and linearly separable under TF-IDF representations.

All models' strong in-domain performance suggests that:

- There is a strong correlation between topic-specific phrases and the dataset's Fake/Real labels.
- For categorization with high accuracy, basic lexical features are adequate.
- Deep models that nearly completely fit for the training distribution include Distil BERT.

The results seem "too good", which emphasizes the necessity for generalization testing because topic leakage could inflate such robust measures.

1.2. Accuracy Comparison

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.007400	0.005315	0.999105	0.999174	1.000000	0.998349
2	0.003000	0.003972	0.999488	0.999528	1.000000	0.999056
3	0.000000	0.005059	0.999488	0.999528	1.000000	0.999056

	Accuracy_Score	F1_Score
SVM Classifier	0.985268	0.995638
Random Forest Classifier	0.994885	0.995296
Logistic regression	0.987852	0.988820
Naive Bayes Classifier	0.946803	0.951151

Figure 3: Accuracy comparison for DistilBERT and classical ML models.

Figure Interpretation:

Distil BERT performs marginally better than all classical ML models. Strong topic-based separability is already demonstrated by the >99% accuracy that classical approaches using TF-IDF can attain.

2. Cross - Topic Generalisation Performance (COVID - 19 Dataset)

Testing whether models trained on political and news content generalize to a new, unexplored domain – COVID-19 articles is the fundamental goal of this project.

We assess Distil BERT, the best model. Training never used the COVID dataset.

2.1. Performance on Unseen COVID Dataset

Exact results from notebook:

Metric	Value
Accuracy	22%
Precision	0.984
Recall	0.213
F1 – Score	0.351

Interpretation

- A significant failure to generalize was evident as accuracy fell from 99.9% --> 22%.
- Precision stayed incredibly high, which means:
 - The model was almost always right when it predicted "Real".

- Recall collapsed, meaning:
 - Most real cases were not identified by the model.
 - The majority has been misclassified.
- These findings show that the model significantly overfits to the training data's topic-specific distributions.

2.2. Cross - Topic Accuracy Drop

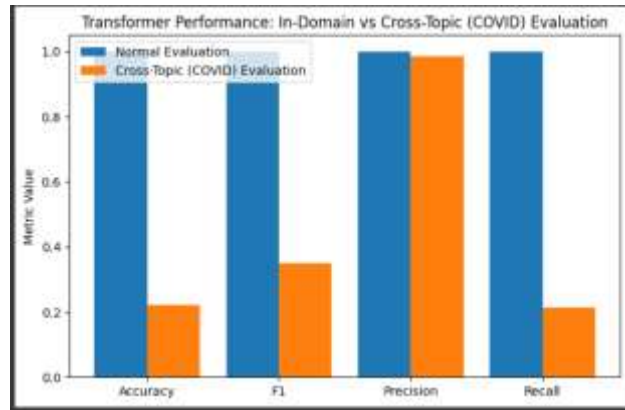


Figure 4: DistilBERT accuracy drop from in - domain test set to COVID dataset.

Interpretation:

The size of the generalization gap is shown graphically in this figure:

- In - domain: 99.94% accuracy
- Cross - topic: 22% accuracy

The performance collapse signifies:

- The model learns topic-specific patterns by heart.
- It does not fully comprehend the semantics of disinformation.
- Distribution shift affects even the most sophisticated transformers.

3. Error Analysis & Topic Leakage

3.1. Topic Leakage

We looked at mistakenly labelled samples and discovered:

- "Politics" keywords were closely linked to both authentic and fraudulent labels.
- models learned to associate:
 - "Biden", "Trump", "White House" with specific classes
 - not the *truthfulness* of the article

This explains their collapse on COVID - 19 news:

- the vocabulary and style are quite different.
- topic cues vanish
- models cannot rely on memorised patterns

3.2. Failure Cases

Typical errors included:

1. Health misinformation phrased neutrally --> **misclassified**
2. Scientific articles --> mislabelled as “Fake” due to unfamiliar terms
3. Short COVID snippets --> Insufficient context for classification

SECTION 5 - LIMITATIONS

This research offers valuable findings regarding how fake-news detection systems perform across different contexts, but specific constraints need to be addressed.

1. Dataset Limitations

The core dataset utilized for distinguishing fake from real news draws predominantly from two subject areas: political coverage and international affairs. Such concentrated subject matter creates an uneven representation that encourages the systems to identify patterns based on word choice and presentation format linked to these particular domains, rather than developing recognition of fundamental misinformation traits. This results in potentially misleading accuracy measurements when testing within similar contexts, as the system simply recognizes familiar terminology and compositional approaches.

Additionally, the source material reflects journalism from an earlier period. Misinformation tactics and communication patterns change quickly, particularly during unprecedented situations like the coronavirus health emergency. The historical nature of the training examples therefore diminishes the system's capacity to identify contemporary forms of false information that employ different linguistic approaches.

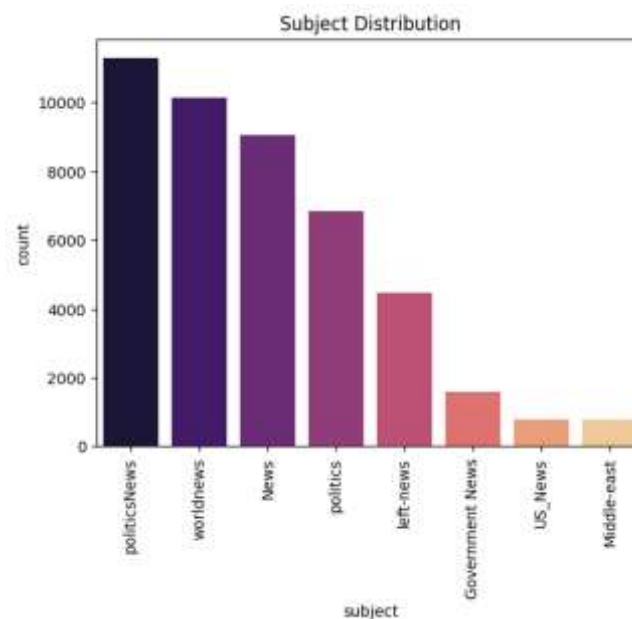


Figure 5: Subject wise article distribution

2. Preprocessing and Representation Limitations

TF - IDF Limitations

The TF-IDF approach quantifies term occurrence rates while failing to understand meaning or contextual relationships. This creates significant vulnerabilities when encountering:

- Shifts in language use
- Changes in subject matter
- Novel words or phrases

Consequently, systems relying on TF-IDF struggle considerably when applied to different contexts than their training environment.

Token Processing Restrictions in Transformer Models

DistilBERT operates with a predetermined input size limitation of 256 tokens maximum. Since numerous news pieces surpass this threshold, this results in:

- Removal of potentially crucial information from longer texts
- Missing subtle details essential for accurate categorization

Alternative transformer architectures designed for extended sequences could potentially resolve this issue, though resource limitations prevented their implementation in this study.

3. Experimental Limitations

The cross-domain assessment relied solely on a single out-of-distribution dataset focused on coronavirus-related content. While this effectively demonstrates performance decline when encountering unfamiliar subject areas, incorporating additional excluded categories (such as economic reporting, athletic coverage, or environmental issues) would provide more robust validation of the findings.

Moreover, traditional machine learning approaches were not applied to the coronavirus dataset because of the significant processing demands associated with creating and analyzing extensive term-frequency representations across disparate domains.

4. Computational Constraints

Transformer architectures demand considerable graphics processing capacity for effective model development. Given available infrastructure limitations:

- The DistilBERT system underwent only three training cycles
- Reduced data batch quantities were necessary
- Systematic optimization of model parameters was not feasible

Access to enhanced computational capabilities might have yielded superior performance or revealed alternative behavioral patterns.

5. Ethical and Interpretability Limitations

Systems designed to identify false information present several potential issues:

- They may perpetuate existing viewpoint-related prejudices
- Authentic reporting could be incorrectly identified as fabricated content
- The reasoning behind categorization outcomes often remains unclear

Analytical tools for examining model logic, including SHAP and LIME frameworks, were excluded from this investigation but might illuminate the underlying decision-making mechanisms.

SECTION 6 - CONCLUSIONS

This project conducted a detailed comparative and generalization study of machine learning models used for misinformation detection. Classical ML models (SVM, Random Forest, Logistic Regression, Naïve Bayes) combined with TF-IDF features, as well as a fine-tuned transformer model (DistilBERT), were evaluated on both in-domain and cross-topic datasets.

The results demonstrate that all models achieved extremely high performance on the in-domain test set, with the SVM and DistilBERT models attaining accuracy levels above 99%. However, when evaluated on an unseen COVID-19 news dataset, DistilBERT which was expected to generalize best experienced a severe performance decline, dropping from 99.94% accuracy to 22% accuracy. Although precision remained high, recall collapsed, revealing a substantial generalization gap. These findings confirm that high in-domain performance does not imply real-world robustness. The models did not learn reliable semantic characteristics of misinformation; instead, they largely memorized topic-specific patterns in the training data. This highlights a critical limitation of current fake-news detection systems and underscores the need for more diverse datasets, domain adaptation techniques, and robust evaluation practices.

Future Directions

1. Use more diverse, multi-topic, and multilingual datasets
2. Implement domain-adversarial training to reduce topic sensitivity
3. Explore longer-context transformer architectures (e.g., Longformer, DeBERTa)
4. Incorporate interpretability tools to understand model decisions
5. Expand cross-topic evaluation to multiple unseen domains

Contribution

This study provides clear empirical evidence that:

1. Traditional metrics overestimate performance
2. Topic leakage inflates accuracy
3. Real-world misinformation detection requires rigorous generalization testing

These insights contribute meaningfully to improving the reliability and fairness of automated misinformation detection systems.

SECTION 7 - REFERENCES

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